

Task3:

configuration	model	accuracy	precision	recall	F1 score
BOW (excluding unseen)	Naive bayes	0.8100	0.8101	0.8100	0.8100
BOW (excluding unseen)	Logistic regression	0.8050	0.8061	0.8051	0.8049
BOW (including unseen)	Naive bayes	0.8025	0.8025	0.8025	0.8025
BOW (including unseen)	Logistic regression	0.8125	0.8147	0.8127	0.8122
BOW (binarized)	Naive bayes	0.8500	0.8500	0.8500	0.8500
BOW (binarized)	Logistic regression	0.8550	0.8551	0.8550	0.8550
Content words only	Naive bayes	0.8100	0.8100	0.8100	0.8100
Content words only	Logistic regression	0.8125	0.8138	0.8127	0.8124
Content words binarized	Naive bayes	0.8475	0.8476	0.8475	0.8475
Content words binarized	Logistic regression	0.8625	0.8625	0.8625	0.8625
BOW (negation)	Naive bayes	0.8225	0.8229	0.8226	0.8225
BOW (negation)	Logistic regression	0.7950	0.8008	0.7953	0.7941
BOW (binarized+negation)	Naive bayes	0.8350	0.8350	0.8350	0.8350
BOW (binarized+negation)	Logistic regression	0.8600	0.8608	0.8601	0.8599

Task 4:-

Lexical Dilution via Metadata and Cast Lists

Both the classifiers consistently flagged a heavily negative review of Star Wars: Episode I as a False Positive in almost all the configurations. One reason for this seems to be the massive block of introductory metadata at the start of the review, like: "director : george lucas cast : liam neeson , ewan mcgregor , natalie portman ... screenplay : george lucas producers : rick mccallum runtime : 131 min". In a Bag-of-Words model, actors and directors who are generally associated with critically acclaimed movies in the training

data can become highly weighted positive features. So, this huge block of neutral and statistically positive proper nouns ends up diluting the actual negative sentiment words that come later in the review. Basically, the model gives too much importance to the metadata and not enough to what the reviewer is actually saying about the movie.

Conditional Praise and Hypothetical Structures

Statistical models also have trouble understanding sentences where something is described positively in a hypothetical or conditional situation, and then that praise is taken back. A Naive Bayes False Positive for *Wide Awake* is a good example: "done well , it could have been a gem among the wasteland of modern children's cinema . unfortunately , it isn't .". The algorithm sees strong positive words like "gem" and "well" and gives them a lot of weight. But it doesn't really understand that "could have been" makes the statement hypothetical, rather than an actual statement that the movie is good. The word "unfortunately" also reverses the meaning of the previous sentence, but it is not a direct logical negator like "not" or "never". Because of this, it also bypasses the Task 2 negation heuristic completely. A human reader can understand the overall meaning pretty easily, but the statistical model mainly sees the positive words.

Domain-Specific Vocabulary and Plot Conflation

Another issue is that reviews often describe gritty, violent, or otherwise negative subject matter even when the reviewer actually has a positive opinion about the movie. Both models generated a False Negative for a heist film review which says: "robert deniro stars as a risk-adverse safecracker who wants to retire from crime... the plot is mostly straightforward suspense with little nonsense .". Words like "crime", "risk-adverse", and "nonsense" are very common in negative reviews, so the classifiers associate them with the negative class. The problem is that the model cannot really distinguish between describing the plot of a dark movie and actually criticising the quality of the movie. So, it ends up penalizing the review just because the review is talking about negative or violent subject matter. This is a problem with relying too much on individual words without understanding the context they are being used in.

Semantic Negation vs. Logical Negation (Sarcasm)

The Task 2 negation rule works reasonably well for explicit logical negators by adding the **NOT_** prefix, but it completely struggles with semantic negation and sarcasm. A False Positive for *Burnt Money* says: "burnt money is the perfect festival film . it will show once or twice , and then no one , thankfully , will ever have to hear from it again ... this

gritty film ... has its only highlight in a well-designed title sequence.". The classifier sees words like "perfect", "highlight", and "well-designed" and strongly associates them with the positive class. But the reviewer is actually making fun of the movie. Saying that it is "perfect" because it will only be shown once or twice and then people will never have to hear about it again is obviously negative when you understand the context. The same thing happens with the "only highlight" being the title sequence. The reviewer is not actually praising the movie. They are using positive words sarcastically to make a negative point. A Bag-of-Words model cant really understand this kind of contextual irony.

Discourse Markers and the "Guilty Pleasure" Concession

Another problem happens with reviews that first spend a lot of time talking about the flaws of a movie, and then give a concession where they admit that they still enjoyed it. A False Negative review says: "the story and characters are so one-dimensional and contrived that you know you shouldn't be enjoying yourself . but somehow , the movie just tries so hard that you begin to forgive it and get into the swing of things.". There are several strongly negative words here, especially "one-dimensional" and "contrived", and these end up overpowering the weaker, more informal positive phrases like "forgive it" and "swing of things". The model basically treats all of these words as an unstructured pile and counts the positive and negative features without really understanding how the sentences are connected. A human reader can recognize that "but somehow" introduces a concession and that the final part is probably closer to the reviewer's actual overall judgement. The classifier doesnt understand this discourse structure, so it ends up predicting the review as negative even though the reviewer ultimately admits that they enjoyed the movie.

Extra credit:

configuration	model	accuracy	precision	recall	F1 score
BOW (polarity lexicon)	Naive bayes	0.8225	0.8226	0.8225	0.8225
BOW (polarity lexicon)	Logistic regression	0.8225	0.8226	0.8219	0.8200